# SUBSCRIPTION MANAGEMENT & CHURN ANALYTICS SYSTEM

---

## DECLARATION

I hereby declare that the project titled **"Subscription Management & Churn Analytics System"** is an original work developed by me.

This project simulates a real-world subscription management platform that tracks customer subscriptions, analyses churn behaviour, and generates revenue analytics using modern backend technologies.

The project demonstrates the practical implementation of **Python, FastAPI, MySQL, and data analytics libraries such as Pandas and NumPy** to build a scalable subscription analytics system.

---

## TABLE OF CONTENTS


1. Executive Summary
2. Introduction
3. Problem Statement
4. Proposed Solution
5. System Architecture
6. Technology Stack
7. Functional Modules
8. Database Design
9. Churn Analytics Logic
10. Workflow of the System
11. Error Handling & Validation
12. Advanced Features
13. Testing Strategy
14. Performance & Reliability
15. Future Enhancements
16. Conclusion
17. GitHub Repository


---

## 1. EXECUTIVE SUMMARY

The **Subscription Management & Churn Analytics System** is a web-based backend application developed to manage customer subscriptions and analyse churn-related metrics.

The system allows organizations to:

- Manage customer subscription records
- Track active and cancelled subscriptions
- Analyse churn rate and revenue metrics
- Identify high-risk customers likely to cancel subscriptions
- Generate subscription analytics insights

By integrating **FastAPI with MySQL and data analytics libraries**, the system provides an efficient platform for managing subscription-based businesses.

---

## 2. INTRODUCTION

Subscription-based businesses require efficient systems to track customer subscriptions and analyse business performance.

Manual subscription tracking can lead to:

• Inefficient customer record management
• Difficulty in analysing churn patterns
• Lack of revenue insights
• Errors in subscription tracking

This project provides a **digital solution for managing subscription data and analysing customer churn using modern backend technologies**.

---

## 3. PROBLEM STATEMENT

Traditional subscription tracking systems often face the following problems:

• Manual management of subscription records
• Lack of automated churn analytics
• Difficulty in tracking revenue metrics
• Limited insights into customer retention

The challenge was to design a system that can **efficiently manage subscription data, calculate churn metrics, and generate business insights using data analytics tools.**

---

## 4. PROPOSED SOLUTION

The proposed system provides a **centralized subscription management platform** that allows organizations to:

• Add and manage customer subscriptions
• Track subscription status and duration
• Analyse churn behaviour using analytics
• Calculate key business metrics such as MRR and ARPU
• Generate data insights for decision making

This system helps businesses improve **customer retention and subscription analytics**.

---

## 5. SYSTEM ARCHITECTURE

**Logical Flow Model**

Admin / Business User
↓
Customer Management Module
↓
Subscription Data Processing
↓
Analytics Engine (Pandas / NumPy)
↓
Churn & Revenue Insights
↓
Reports & Analytics Dashboard

---

**Architecture Layers**

**Presentation Layer**

Handles API requests and responses using **FastAPI** and displays data through frontend templates.

**Business Logic Layer**

Processes customer subscription data and performs churn analytics.

**Data Layer**

Stores customer and subscription records in the **MySQL database**.

---

**6. TECHNOLOGY STACK**

**CORE STACK**

Backend: Python, FastAPI

Frontend: HTML, CSS, Jinga2 Templates

Database: MySQL

Data Analytics: Pandas, NumPy

ORM: SQLAlchemy

API Documentation: Swagger UI (FastAPI Docs)

Development Tool: VS Code

Version Control: Git & GitHub

Server: Uvicorn

---

**PYTHON CONCEPTS USED**

Functions

Classes & Object-Oriented Programming

API Development

Database Integration

Exception Handling

Data Processing with Pandas

---

**7. FUNCTIONAL MODULES**

**CUSTOMER MANAGEMENT MODULE**

• Add new customers
• Store customer information such as name, email, and subscription details

• Update customer records
• Delete customer data

---

**SUBSCRIPTION MANAGEMENT MODULE**

• Manage subscription plans
• Track subscription status (Active / Cancelled)
• Store subscription starts and end dates

---

**CHURN ANALYTICS MODULE**

• Calculate churn rate
• Identify customers likely to churn
• Analyse customer retention patterns

---

**REVENUE ANALYTICS MODULE**

• Calculate Monthly Recurring Revenue (MRR)
• Calculate Average Revenue Per User (ARPU)
• Analyse revenue trends

---

**DASHBOARD ANALYTICS MODULE**

• Display subscription statistics
• Show churn and revenue insights

---

**8. DATABASE DESIGN**

**Customer Table**

Stores customer information.

Fields include:

Customer ID
Customer Name
Email
Subscription Plan
Subscription Status
Subscription Start Date

---

**Subscription Table**

Stores subscription-related details.

Fields include:

Subscription ID
Customer ID
Plan Type

Monthly Fee
Subscription Status

---

**Analytics Table (Optional)**

Stores processed analytics results if required.

Fields include:

Analytics ID
Churn Rate
Monthly Revenue
Average Revenue

---

**9. CHURN ANALYTICS LOGIC**

The system uses **Pandas and NumPy libraries** for churn analytics.

Analytics include:

• Churn rate calculation
• Monthly recurring revenue (MRR)
• Average revenue per user (ARPU)
• Customer retention statistics

These analytics help organizations understand customer behaviour and improve retention strategies.

---

**10. WORKFLOW OF THE SYSTEM**

Step 1 → Start the FastAPI application

Step 2 → Connect to MySQL database

Step 3 → Add customer subscription data

Step 4 → Store subscription details in database

Step 5 → Process data using Pandas

Step 6 → Calculate churn and revenue metrics

Step 7 → Generate analytics insights

Step 8 → Display results via API responses

---

**11. ERROR HANDLING & VALIDATION**

The system includes validation mechanisms to maintain data accuracy.

Features include:

• Input validation for customer data
• Database error handling
• API response validation
• Data verification for subscription records

This ensures **stable and reliable system performance**.

---

**12. ADVANCED FEATURES**

• Churn analytics using Pandas

• Subscription revenue insights

• REST API architecture using FastAPI

• Automatic API documentation via Swagger

• Scalable modular backend architecture

---

**13. TESTING STRATEGY**

The system was tested using the following methods:

• API endpoint testing

• Database operation testing

• Churn analytics accuracy testing

• Input validation testing

• Performance testing

---

**14. PERFORMANCE & RELIABILITY**

The system provides:

• Fast API response time

• Efficient database queries

• Accurate analytics calculations

• Scalable backend architecture

The system can be extended to support **large-scale subscription datasets**.

---

**15. FUTURE ENHANCEMENTS**

• Machine learning based churn prediction

• Email notification for subscription expiry

• Payment gateway integration

• Interactive analytics dashboard

• Role-based authentication system

---

## 16. CONCLUSION

The **Subscription Management & Churn Analytics System** successfully demonstrates the implementation of a backend platform for managing subscription-based businesses.

The project highlights:

• Efficient customer subscription management

• Churn analytics using Pandas and NumPy

• REST API development using FastAPI

• Integration with MySQL database

This project provides a strong foundation for building **advanced subscription analytics platforms for SaaS businesses**.

---

## GITHUB REPOSITORY

Click the link below:

Subscription Management and Churn Analytics System